Intelligent System
Automatic License Plate Recognition

Luca Granucci

Lorenzo Marranini

DIPARTIMENTO DI INGEGNERIA DELL'INFORMAZIONE

UNIVERSITÀ DI PISA

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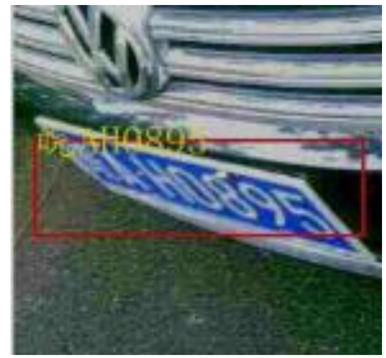
The Problem

License plates captured in real-world scenarios are highly subject to environmental noise, changing lighting conditions, and severe geometric deformations.




Our Goal: Design and evaluate an End-to-End Automatic License Plate Recognition pipeline that can be used in real applications as:

- Highway Free-Flow Tolling Systems
- Smart Parking & Smart Access Gates
- Automated Traffic Surveillance

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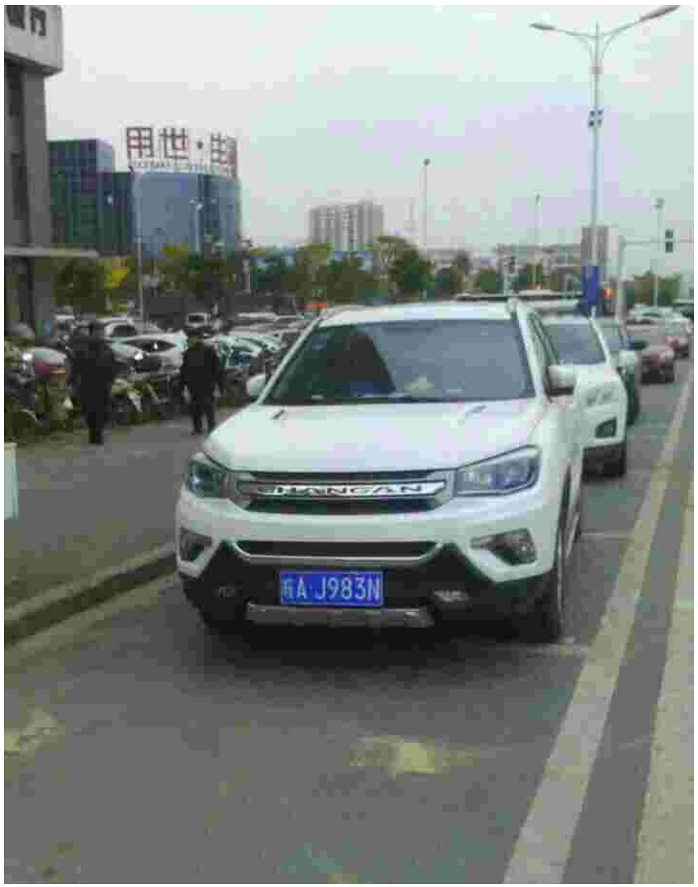
The Task

Input



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Task 1: Plate Recognition

Task 2: Character Recognition

Final Output: [皖AJ983N]

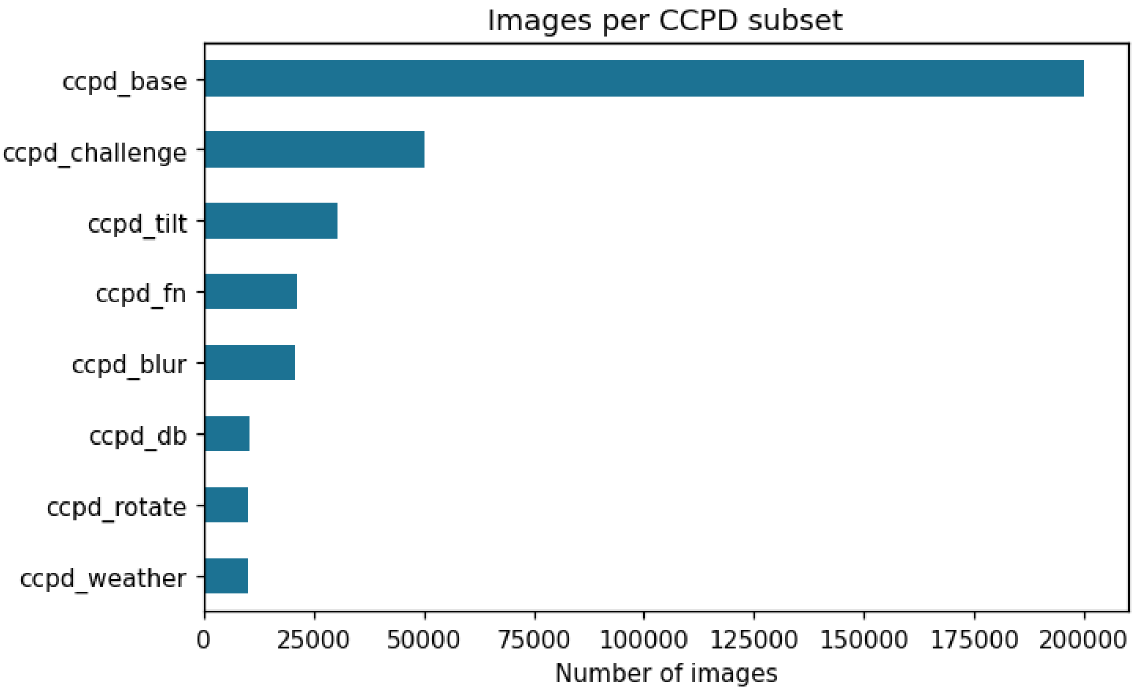
The Dataset

CCPD (Chinese City Parking Dataset, ECCV)

250K+ Images labeled with:

- Bounding box of license plate
- License plate characters
- License plate corners key points

Grouped by positions and conditions



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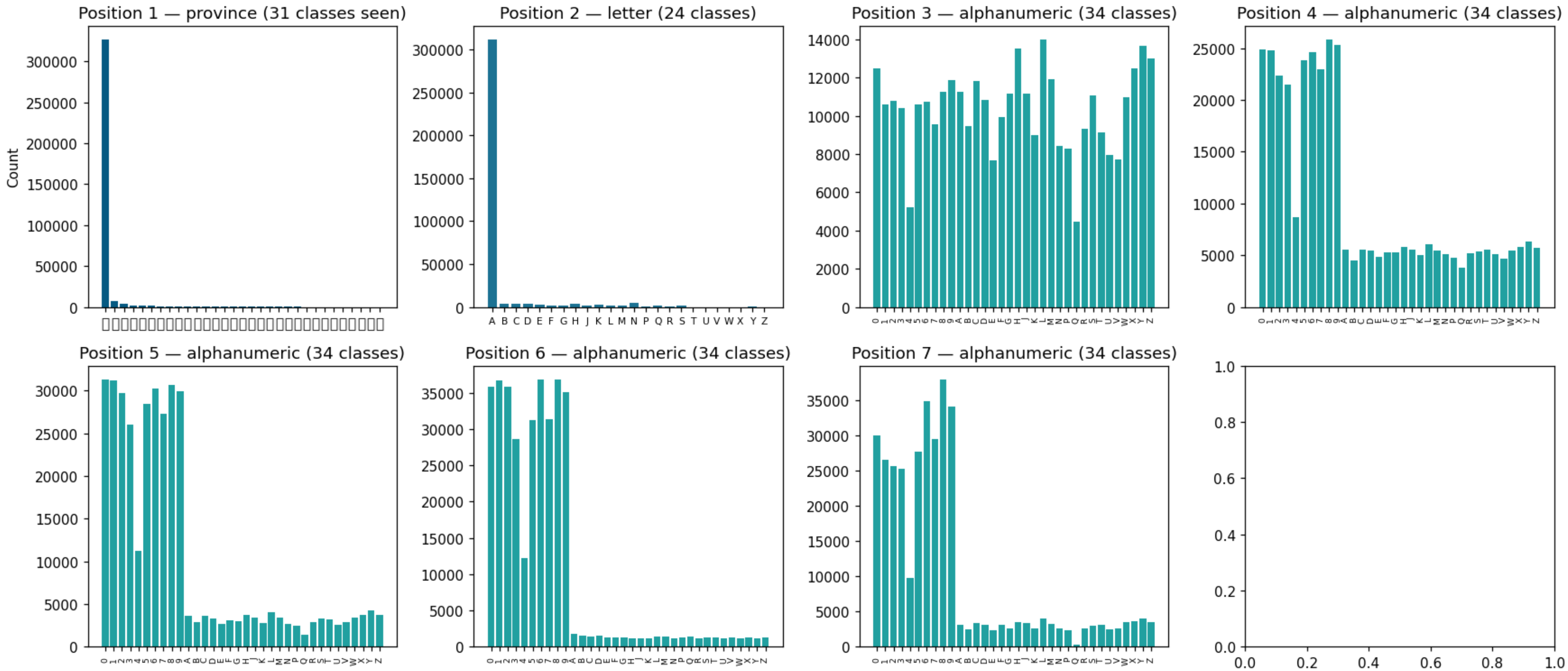
Dataset Exploration

License Plate with 7 alphanumeric characters



Computer Vision Models

Task 1: plate localization

- Yolo11pose
- Yolo11seg

Task 2: plate characters recognition

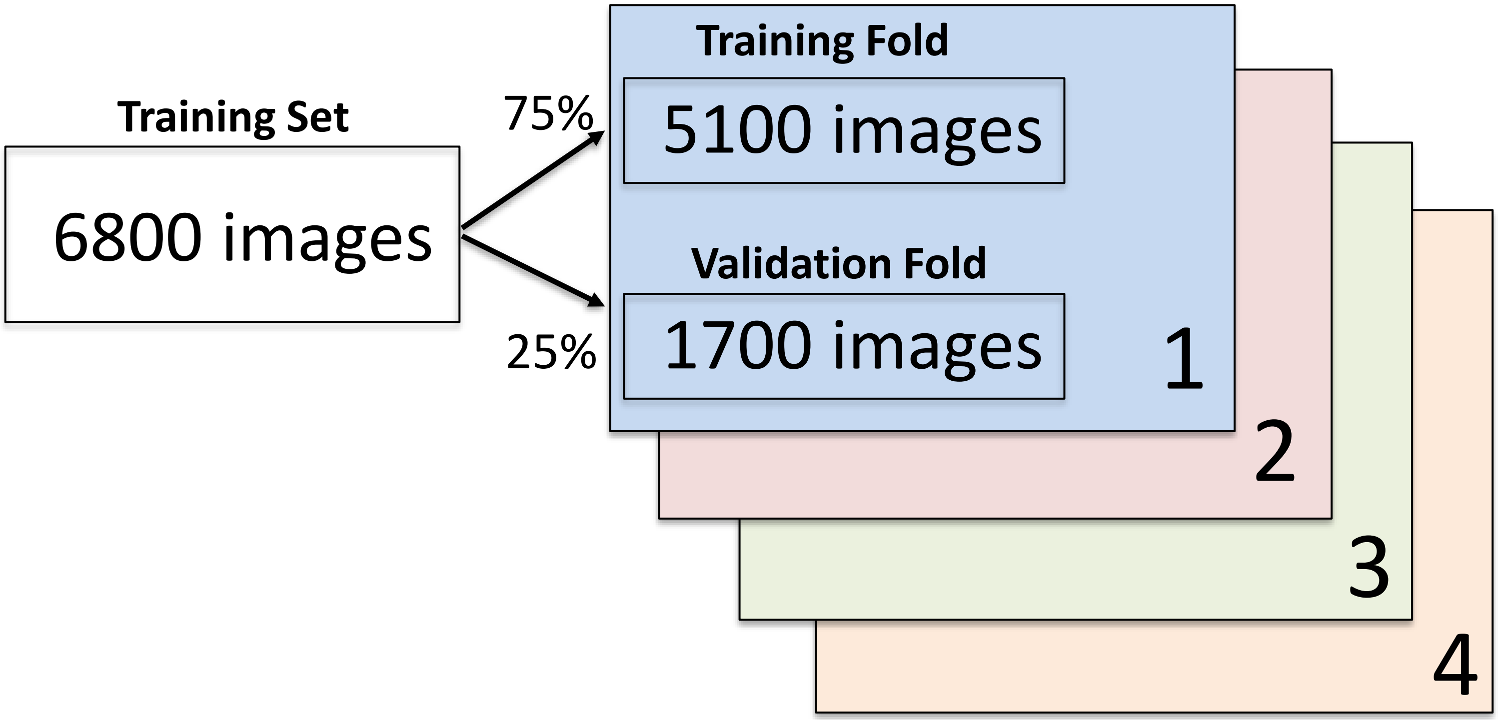
- PaddleOCR
- LPRNET
- CNN

Setup & Training Architecture:

Nvidia GeForce RTX 2060, 6GB VRAM

Yolo Cross Validation

Dataset subsample of 6800 images



Stratification preserved the relative proportion of samples from each CCPD subset in every fold.

Plate Localization

Yolo11pose

Yolo11seg

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Plate Localization Results

Model	Pretrained	Size	mAP@50-95	Precision	Recall	Mean Corner Error
Pose	Yes	n	0.99351	0.99883	0.99736	4.22 px
Pose	Yes	s	0.99426	0.99826	0.99876	3.79 px
Pose	Yes	m	0.99366	0.99902	0.99850	4.19 px
Seg	Yes	n	0.88685	0.99563	0.995528	7.44 px
Seg	Yes	s	0.88301	0.99816	0.996322	7.93 px
Seg	Yes	m	0.88685	0.99831	0.996188	8.06 px

Plate Warping

Following the first step, a perspective transformation was required to warp the license plate image, ensuring optimal recognition by the Step 2 computer vision models.

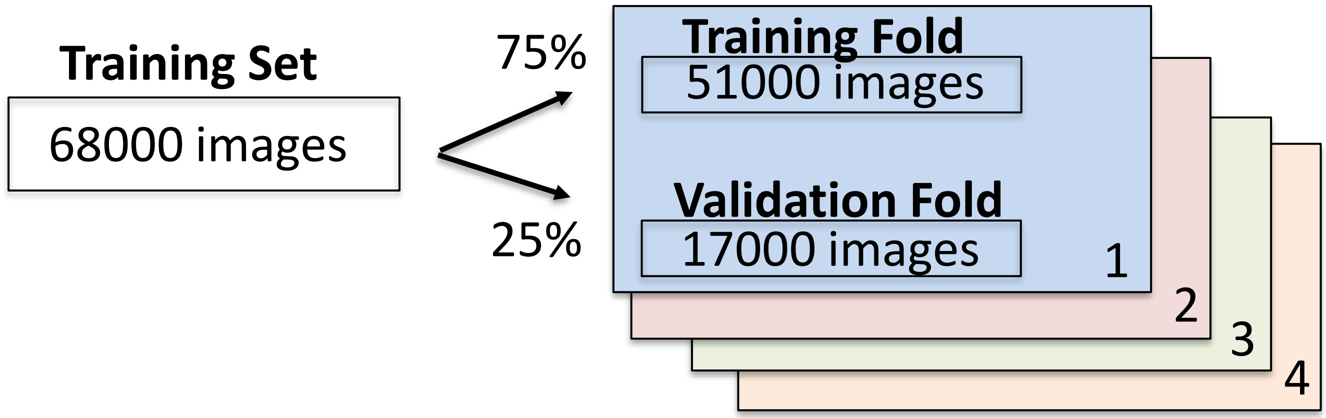
Bounding Box

Homography Process

Warped output

Character Recognition Results

Dataset subsample of 68000 images



Model	Full plate accuracy	Mean edit distance
PaddleOCR	0.717	0.569
7-head CNN	0.463	1.004
LPRNet	0.963	0.049

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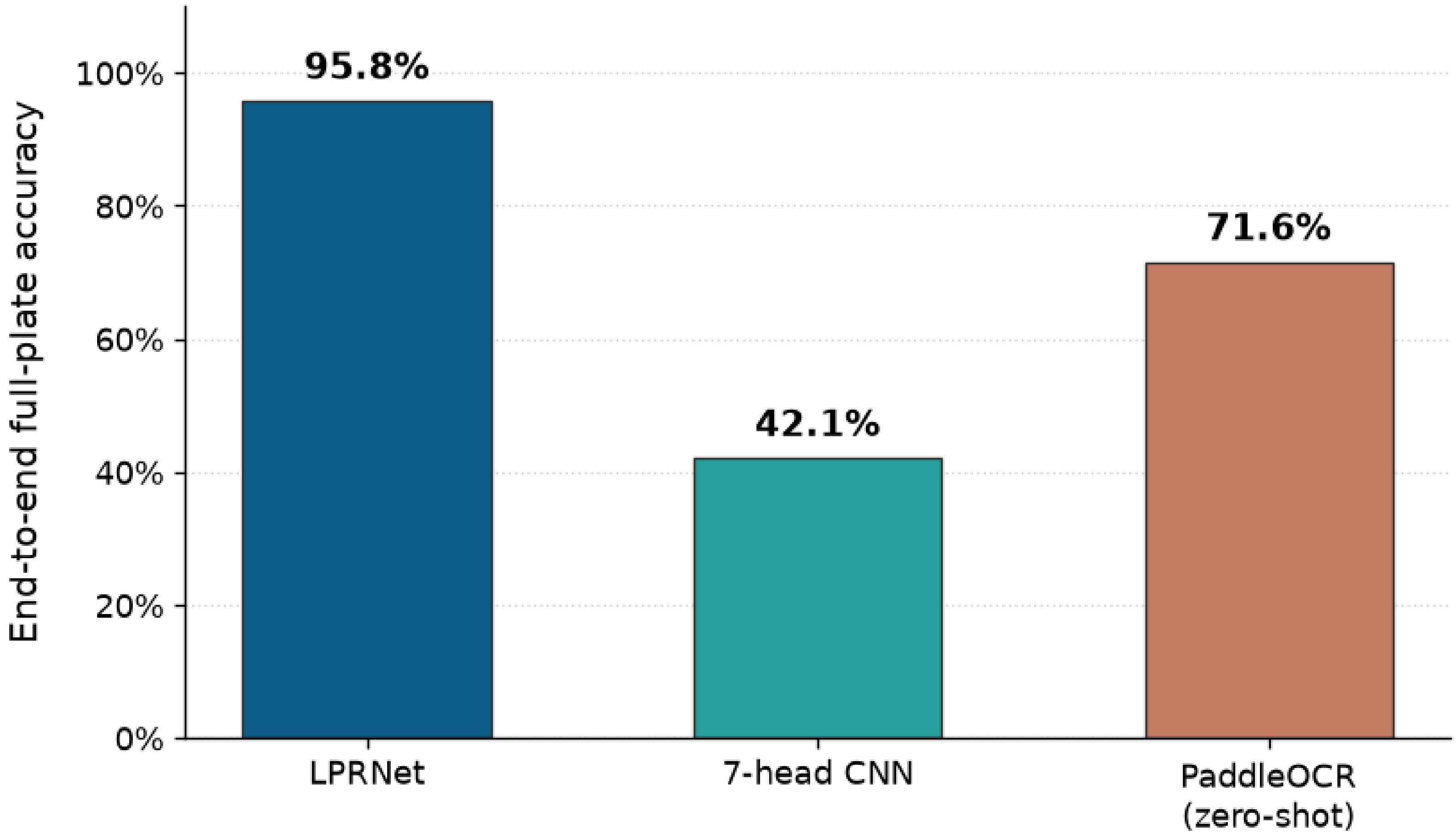
End-to-End Results

Tested on 5000 unseen images



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Simulation Scenario

We tested our end-to-end pipeline on a real-world video from a dash cam

Conclusions

In conclusion, the best-performing pipeline is the combination of YOLO11 Pose Small and LPRNet.
This specific architecture delivered the highest performance across both tasks:

- **Plate Localization:** The YOLO11 Pose Small model achieved the best detection results. It recorded the highest mAP@50-95 at 0.99426 and the lowest absolute corner error of 3.79 pixels.

- **Character Recognition:** In the character recognition task, LPRNet significantly outperformed alternative models like the 7-head CNN and PaddleOCR. Driven by this model, our complete pipeline reached an outstanding end-to-end full-plate accuracy of 95.8% on 5,000 unseen test images.

Future Work

Possible Improvements:

- Extending the system to handle different license plate formats and countries
- Model compression techniques such as quantization
- Enhancing performance in low-visibility conditions
- Integration with surveillance and smart city systems